



Telco

Offer, Load & Transaction Analysis

- By Nissan





Project Overview

This project focuses on analyzing Telco offer, load & transaction using SQL to gain insights into product sales, revenue, customer behavior & operational performance. Key analysis include identifying top-selling products by bundle type, calculating total revenue and average prices, and evaluating cancellation & complaint rates. Advanced SQL techniques such as CTE and Window functions are used in this project. This project provides strategic recommendations to optimize telecom offerings, improve sales performance, and enhance customer satisfaction.





Project Objective

- Analyze telecom transactions for insights in sales and customer behavior.
- Identify top products by bundle type and calculate revenue metrics.
- Assess cancellation, complaint rates and operational efficiency.
- Use Advanced techniques for performance analysis.
- Provide Recommendations to optimize sales and enhance satisfaction.





Which Product is most ordered per bundle type?

```
WITH BOX AS
(SELECT
BUNDLE_TYPE,
CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
COUNT(ORDER_ID)AS TOTAL_ORDER,
ROW_NUMBER()OVER (PARTITION BY BUNDLE_TYPE ORDER BY COUNT(ORDER_ID)DESC)AS RANKS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1,2)
SELECT * EXCEPT(RANKS)
FROM BOX WHERE RANKS=1;
```

BUNDLE_TYPE	PRODUCT_NAME	TOTAL_ORDER
Internet	ROBI-10 GB	1298
Intenet & Minute	GP-40 GB - 800 min	581
Minute	GP-500 min	666



The analysis identified the most ordered products per bundle type from the dataset. The top product in the Internet category is ROBI-10 GB, with 1298 orders. On the Internet & Minute category, GP-40 GB - 800 min leads with 581 orders, while the top voice package is GP-500 min, totaling 666 orders. This insight highlights consumer preferences, indicating a strong demand for data services alongside voice communication options.



Which Bundle Type is least ordered in each Division?

```
WITH Y AS
(SELECT
BUNDLE_TYPE,
DIVISION,
CONCAT(UPPER(OPERATOR), "- ", PRODUCT_NAME)AS PRODUCT_NAME,
COUNT(ORDER_ID)AS TOTAL_ORDER,
ROW_NUMBER() OVER ( PARTITION BY DIVISION ORDER BY COUNT(ORDER_ID) DESC)AS RANKS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1,2,3)
SELECT * EXCEPT(RANKS)
FROM Y
WHERE RANKS = 1;
```

BUNDLE_TYPE	DIVISION	PRODUCT_NAME	TOTAL_ORDER
Internet	Barisal	ROBI- 20 GB	53
Minute	Mymensingh	GP- 500 min	78
Internet	Chittagong	ROBI- 10 GB	609
Internet	Dhaka	ROBI- 10 GB	342
Internet	Sylhet	ROBI- 10 GB	81
Minute	Rajshahi	GP- 500 min	87
Minute	Rangpur	GP- 500 min	95
Intenet & Minute	Khulna	GP- 40 GB - 800 min	66



The analysis identifies least ordered bundle types in each division. In Barisal, ROBI-20GB has 53 orders while in Mymensingh, GP-500MIN has 78 order. In Chattogram, ROBI-10GB leads with 609 orders while in Dhaka ROBI-10GB leads with 342 order. Sylhet's least ordered product is also ROBI-10GB which has only 81 orders and Rajshahi is GP-500MIN with 87 order. In Rangpur least order is 95 for GP-500MIN product & Khulna with 66 order for GP-40GB-800MIN package.



What is the total revenue and average selling price per product, bundle type, operator and division.

```
SELECT
    DIVISION,
    OPERATOR,
    SUM(CASE WHEN BUNDLE_TYPE = 'Internet' THEN SELLING_PRICE ELSE 0 END) AS TOTAL_REVENUE_INTERNET,
    AVG(CASE WHEN BUNDLE_TYPE = 'Internet' THEN SELLING_PRICE ELSE 0 END) AS AVG_SELLING_PRICE_INTERNET,
    SUM(CASE WHEN BUNDLE_TYPE = 'Intenet & Minute' THEN SELLING_PRICE ELSE 0 END) AS TOTAL_REVENUE_INTERNET_AND_MINUTE,
    AVG(CASE WHEN BUNDLE_TYPE = 'Intenet & Minute' THEN SELLING_PRICE ELSE 0 END) AS AVG_SELLING_PRICE_INTERNET_AND_MINUTE,
    SUM(CASE WHEN BUNDLE_TYPE = 'Minute' THEN SELLING_PRICE ELSE 0 END) AS TOTAL_REVENUE_MINUTE,
    AVG(CASE WHEN BUNDLE_TYPE = 'Minute' THEN SELLING_PRICE ELSE 0 END) AS AVG_SELLING_PRICE_MINUTE
FROM
    `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
WHERE
    DIVISION IS NOT NULL
GROUP BY
    1, 2;
```

DIVISION	OPERATOR	TOTAL_REVENUE	II AVG_SELLING	I TOTAL_REVENUE	AVG_SELLING	TOTAL_REVENUE	AVG_SELLING
Rangpur	robi	64075	113.608156	122103	216.4946809	24711	43.81382979
Rangpur	gp	57367	69.20024125	278171	335.5500603	44914	54.17852835
Rangpur	airtel	37618	86.08237986	123604	282.8466819	10752	24.60411899
Rangpur	bl	36908	178.2995169	39175	189.2512077	9499	45.88888889
Khulna	gp	20713	50.39659367	145083	353	23312	56.72019465
Khulna	airtel	24774	102.3719008	69108	285.5702479	3240	13.38842975
Chittagong	robi	457600	169.2307692	280930	103.8942308	91523	33.84726331
Chittagong	airtel	185087	124.3864247	326999	219.7573925	20323	13.65793011
Chittagong	gp	90515	126.7717087	220501	308.82493	25641	35.91176471
Sylhet	gp	62494	154.6881188	86045	212.9826733	21915	54.2450495
Sylhet	airtel	58403	128.3582418	116574	256.2065934	5012	11.01538462
Dhaka	bl	136817	136.953954	216298	216.5145145	76156	76.23223223
Rajshahi	gp	47385	75.21428571	194711	309.0650794	38553	61.1952381
Rajshahi	robi	73948	203.7134986	44569	122.7796143	7074	19.48760331
Rajshahi	bl	21459	137.5576923	40110	257.1153846	5485	35.16025641
Rajshahi	airtel	44939	132.5634218	73068	215.539823	3461	10.20943953
Dhaka	robi	232179	122.3927254	348402	183.6594623	100454	52.95413811
Dhaka	airtel	136663	81.29863177	494810	294.3545509	36798	21.89054134
Dhaka	gp	133690	80.48765804	526057	316.7110175	104787	63.08669476
Barisal	gp	45071	105.5526932	117897	276.1053864	27380	64.12177986
Mymensingh	bl	36416	120.9833887	67177	223.179402	19869	66.00996678
Khulna	robi	26285	89.10169492	55662	188.6847458	22224	75.33559322
Khulna	bl	43755	149.334471	63824	217.8293515	20561	70.17406143
Sylhet	bl	8003	103.9350649	29066	377.4805195	1761	22.87012987
Sylhet	robi	74882	198.1005291	54448	144.042328	5076	13.42857143
Barisal	robi	43060	146.9624573	62490	213.2764505	11002	37.54948805



This indicates trend across regions. Operators like ROBI & GP dominate certain divisions, while others show distinct revenue pattern. The pricing strategy also differ, with some divisions showing higher average selling price per bundle type.



Find the Product and Bundle Type Contributing the Highest Percentage of total Revenue?

```
WITH X AS (  
  SELECT  
    BUNDLE_TYPE,  
    CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,  
    SUM(SELLING_PRICE) AS TOTAL_REVENUE,  
    ROUND(SUM(SELLING_PRICE) / (SELECT  
      SUM(SELLING_PRICE)  
    FROM `organic-reef-427307-i3.offer_load.Offer_load_Transaction`) * 100, 2) AS PERCENTAGE,  
    ROW_NUMBER() OVER (ORDER BY SUM(SELLING_PRICE) DESC) AS RANKS  
  FROM  
    `organic-reef-427307-i3.offer_load.Offer_load_Transaction`  
  GROUP BY  
    BUNDLE_TYPE, OPERATOR, PRODUCT_NAME  
)  
SELECT *  
FROM X  
WHERE RANKS = 1;
```

BUNDLE_TYPE	PRODUCT_NAME	TOTAL_REVENUE	PERCENTAGE	RANKS
Internet & Minute	GP-40 GB - 800 min	327044	3.99	1



The product GP-40GB-800MIN on the Internet & Minute bundle type contributes the highest percentage on the total revenue accounting for 3.99%.



1. Which product had the highest number of sales and what was the total revenue for that product?
2. What is the Maximum and Minimum amount of Discount and commission per product?

1

```
SELECT
CONCAT (UPPER(OPERATOR), "-", PRODUCT_NAME)AS PRODUCT_NAME,
COUNT(ORDER_ID)AS COUNTS,
SUM(SELLING_PRICE)AS TOTAL_REVENUE
FROM
`organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY 1
ORDER BY COUNTS DESC
LIMIT 1;
```

PRODUCT_NAME	COUNTS	TOTAL_REVENUE
ROBI-10 GB	1298	212266

2

```
SELECT
CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME)AS PRODUCT_NAME,
MAX(commission)AS MAX_COMMISSION,
MIN(commission)AS MIN_COMMISSION,
MAX(discount)AS MAX_DISCOUNT,
MIN(discount)AS MIN_DISCOUNT
FROM
`organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY 1;
```

PRODUCT_NAME	MAX_COMMISSION	MIN_COMMISSION	MAX_DISCOUNT	MIN_DISCOUNT
GP-12 GB	10	5	19	14
ROBI-12 GB - 400 min	90	3	69	7
GP-512 MB - 500 min	35	-2	34	10
AIRTEL-5 GB - 300 min	23	15	49	49
GP-110 min	5	-32	2	2
GP-25 GB	173	-75	425	5
GP-500 min	46	-20	37	2
BL-510 min	25	-13	32	2
GP-512 MB - 330 min	35	5	13	5
AIRTEL-5 GB - 535 min	28	0	35	14
AIRTEL-3 GB - 540 min	15	10	19	19
GP-400 min	32	2	17	8
GP-40/50 GB - 1000 min	40	-5	293	208
GP-35 GB - 800 min	76	-20	269	75
AIRTEL-50 GB - 1100 min	60	2	178	103
GP-40 GB - 800 min	100	-47	339	57
GP-1 GB - 1000 min	87	-52	204	49
GP-40 GB - 750 min	75	0	249	99
GP-320 min	15	-9	17	7
ROBI-10 GB	80	-5	149	



The product ROBI-10GB had the highest number of sales which was ordered 1298 times generating a profit of 212266 TK.



The maximum and minimum discount and commission per product will vary depending on the product's pricing and sales performance. Typically, products with higher sales volumes or premium pricing may have higher discounts and commissions.



- 1.Which Operator, Division, offers the Highest avg discount across Products?
- 2.Which customer has the highest number of orders or total spending?

1

```
SELECT
UPPER(OPERATOR) AS OPERATOR,
UPPER(DIVISION) AS DIVISION,
ROUND(AVG(DISCOUNT),2)AS AVG_DISCOUNT
FROM
`organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY
1,2
ORDER BY AVG_DISCOUNT DESC
LIMIT 1;
```

OPERATOR	DIVISION	AVG_DISCOUNT
GP	RANGPUR	111.38

2

```
SELECT
CUSTOMER_ID,
COUNT(ORDER_ID) AS TOTAL_ORDERS,
SUM(SELLING_PRICE)AS TOTAL_SPENDING
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
1
order by
TOTAL_ORDERS DESC
LIMIT
1;
```

CUSTOMER_ID	TOTAL_ORDERS	TOTAL_SPENDING
3011	335	103474



The operator GP in the Rangpur division offers the highest average discount across products, with an average discount of 111.38. This indicates significant discounting strategies employed in that region for GP products.



Customer 3011 has the highest number of orders, with a total of 335 orders. Their total spending amounts to 103,474.



What is the avg purchase frequency of customers and and how does it vary by bundle type?

```
WITH CTE AS
(SELECT
BUNDLE_TYPE,
CUSTOMER_ID,
COUNT(ORDER_ID) PURCHASE_FREQUENCY
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY 1,2)
SELECT
BUNDLE_TYPE,
CASE WHEN CTE.PURCHASE_FREQUENCY BETWEEN 1 AND 5 THEN '1-5'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 6 AND 10 THEN '6-10'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 11 AND 15 THEN '11-15'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 16 AND 20 THEN '16-20'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 21 AND 25 THEN '21-25'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 26 AND 30 THEN '26-30'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 31 AND 35 THEN '31-35'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 36 AND 40 THEN '36-40'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 41 AND 45 THEN '41-45'
WHEN CTE.PURCHASE_FREQUENCY BETWEEN 46 AND 50 THEN '46-50'
ELSE 'ABOVE 50' END AS FREQUENCY_BIN,
COUNT(DISTINCT CUSTOMER_ID) AS COUNT,
ROUND(AVG(PURCHASE_FREQUENCY),2) AS AVG_PURCHASE
FROM
CTE
GROUP BY 1,2;
```



The average purchase frequency of customers reflects how often they buy, varying significantly by bundle type. Combined bundles like Internet & Minute tend to have higher purchase frequencies compared to standalone Internet or Minute bundles, indicating greater customer interest in comprehensive offerings.

BUNDLE_TYPE	FREQUENCY_BIN	COUNT	AVG_PURCHASE
Intenet & Minute	11-15	101	12.74
Intenet & Minute	1-5	1044	2.17
Intenet & Minute	ABOVE 50	15	84.87
Intenet & Minute	21-25	25	23.04
Intenet & Minute	6-10	193	7.69
Intenet & Minute	16-20	45	18
Intenet & Minute	36-40	5	38
Intenet & Minute	41-45	9	42.89
Intenet & Minute	26-30	19	27.63
Intenet & Minute	31-35	12	32.5
Intenet & Minute	46-50	6	48.33
Internet	1-5	1065	1.95
Internet	6-10	163	7.61
Internet	16-20	25	18
Internet	11-15	57	12.75
Internet	26-30	12	27.92
Internet	ABOVE 50	17	100.53
Internet	21-25	15	22.33
Internet	41-45	2	42.5
Internet	46-50	8	47.25
Internet	36-40	7	37
Internet	31-35	6	33.67
Minute	6-10	86	7.72
Minute	1-5	741	1.94
Minute	11-15	34	12.68
Minute	21-25	9	23
Minute	16-20	10	17.2
Minute	ABOVE 50	3	64.33
Minute	41-45	2	42.89



1. During which time of the day, most orders are placed?

2. What is the average fulfillment or delivery time between order time and delivery for each operator and division?

```
SELECT
  EXTRACT(HOUR FROM ORDER_TIME) AS ORDER_HOUR,
  COUNT(ORDER_ID) AS TOTAL_ORDERS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
  1
ORDER BY
  TOTAL_ORDERS DESC
LIMIT 1;
```

ORDER_HOUR	TOTAL_ORDERS
19	2129

```
SELECT
  UPPER(OPERATOR) AS OPERATOR,
  ROUND(AVG(DATE_DIFF(DELIVERY_TIME, ORDER_TIME, MINUTE)), 2) AS DELIVERY_TIME
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
  ORDER_STATUS = "Confirm"
GROUP BY
  1;
```

OPERATOR	DELIVERY_TIME
GP	42.88
ROBI	37.76
AIRTEL	10.28
BL	12.03



Most orders are placed during 19:00 (7 PM), with a total of 2,129 orders. This suggests peak customer activity in the evening hours.



The average delivery time varies by operator, with GP having the longest at 42.88 hours and Airtel the shortest at 10.28 hours. ROBI and BL have delivery times of 37.76 hours and 12.03 hours, respectively. This highlights differences in operational efficiency across operators.





1. Calculate the average delivery time for orders in each division.
2. How many orders were placed in each division?

1

```
SELECT
UPPER(DIVISION),
ROUND(AVG(DATE_DIFF(DELIVERY_TIME, ORDER_TIME, MINUTE)),2)AS DELIVERY_TIME
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
ORDER_STATUS = 'Confirm'
GROUP BY
1;
```

DIVISION	DELIVERY_TIME
RANGPUR	99.83
KHULNA	10.22
CHITTAGONG	45.89
SYLHET	10.33
DHAKA	11.44
RAJSHAHI	10.29
BARISAL	12.23
MYMENSINGH	9.81

2

```
SELECT
DIVISION,
COUNT(ORDER_ID)AS TOTAL_ORDERS
FROM
organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE DIVISION IS NOT NULL
GROUP BY
1;
```

DIVISION	TOTAL_ORDERS
Rangpur	2037
Khulna	1241
Chittagong	5171
Sylhet	1314
Dhaka	6238
Rajshahi	1488
Barisal	1236
Mymensingh	1910



The average delivery time varies significantly across divisions, with Rangpur having the longest at 99.83 hours, while Mymensingh is the fastest at 9.81 hours. Other divisions, like Khulna and Sylhet, also show efficient delivery times around 10-12 hours, highlighting regional disparities in order fulfillment.



The highest number of orders were placed in Dhaka with 6,238 orders, followed by Chittagong with 5,171 orders. Barisal and Khulna had the least orders at 1,236 and 1,241, respectively, showing a significant variation in order volume across divisions.



What are the cancellation rate and complaint rate per operator, and division?

```
WITH X AS
(SELECT
  DIVISION,
  OPERATOR,
  COUNT(ORDER_ID) AS TOTAL_ORDERS,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Cancel" THEN ORDER_ID ELSE NULL END) AS CANCELLED_ORDERS,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Complain" THEN ORDER_ID ELSE NULL END) AS COMPLAINT_ORDERS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE DIVISION IS NOT NULL
GROUP BY
  1,2)
SELECT
  DIVISION,
  OPERATOR,
  ROUND(CANCELLED_ORDERS/ X.TOTAL_ORDERS* 100,2) AS CANCEL_PERCCENTAGE,
  ROUND(COMPLAINT_ORDERS/ X.TOTAL_ORDERS* 100,2) AS COMPLAINT_PERCENTAGE
FROM
  X;
```

DIVISION	OPERATOR	CANCEL_PERCCENTAGE	COMPLAINT_PERCENTAGE
Rangpur	robi	13.65	6.91
Rangpur	gp	12.06	9.53
Rangpur	airtel	10.07	7.78
Rangpur	bl	5.8	8.21
Khulna	gp	9.25	10.22
Khulna	airtel	7.85	7.44
Chittagong	robi	18.23	8.47
Chittagong	airtel	10.22	9.68
Chittagong	gp	14.01	7.84
Sylhet	gp	10.89	5.94
Sylhet	airtel	12.97	6.59
Dhaka	bl	12.01	9.31
Rajshahi	gp	10.63	8.57
Rajshahi	robi	15.15	10.47
Rajshahi	bl	12.18	9.62
Rajshahi	airtel	11.8	9.14
Dhaka	robi	15.87	7.06
Dhaka	airtel	13.03	6.9
Dhaka	gp	15.35	9.03
Barisal	gp	9.13	9.84
Mymensingh	bl	9.97	12.29
Khulna	robi	16.61	8.47
Khulna	bl	13.65	10.92
Sylhet	bl	10.39	18.18
Sylhet	robi	15.87	7.41
Barisal	robi	17.06	9.22
Barisal	bl	13.44	10.67
Mymensingh	airtel	9.17	9.66
Barisal	airtel	12.17	9.13
Mymensingh	gp	14.03	7.51
Mymensingh	robi	14.84	8.94
Chittagong	bl	10.57	14.34



The cancellation and complaint rates vary by operator and division, reflecting differences in service quality. Operators and divisions with higher rates may need to address operational inefficiencies and customer satisfaction issues.



- 1.What are the most common reasons for complaints?
- 2.Which division has the highest number of total sales and transactions?

1

```
SELECT
  REASON,
  COUNT(REASON) AS MOST_COMMON_REASON
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
  1
ORDER BY MOST_COMMON_REASON DESC
LIMIT
  1;
```

REASON	MOST_COMMON_REASON
আপনার নাম্বারে এই অফারটি নেই	1651

2

```
SELECT
  DIVISION,
  SUM(SELLING_PRICE) AS TOTAL_SALES_VALUE,
  COUNT(ORDER_ID) AS TOTAL_TRANSACTIONS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
  DIVISION
ORDER BY
  TOTAL_SALES_VALUE DESC
LIMIT 1;
```

DIVISION	TOTAL_SALES_VALUE	TOTAL_TRANSACTIONS
Dhaka	2543111	6238



The most common complaint is “Apner Number a eei Offer ti nei”, reported 1,651 times. This highlights customer dissatisfaction with offer availability on specific numbers.



Dhaka leads with the highest total sales value of 2,543,111 and the most transactions at 6,238. This indicates Dhaka's dominance in sales and customer activity.



What is the percentage of order status (e.g., completed, canceled) per operator and division?

```
WITH Y AS
(SELECT
  DIVISION,
  OPERATOR,
  COUNT(ORDER_ID) AS TOTAL_ORDERS,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Confirm" THEN ORDER_ID ELSE NULL END ) AS TOTAL_CONFIRMED,
  COUNT( DISTINCT CASE WHEN ORDER_STATUS = "Cancel" THEN ORDER_ID ELSE NULL END) AS TOTAL_CANCELED
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
  DIVISION IS NOT NULL
GROUP BY
  1,2)
SELECT
  UPPER(DIVISION) AS DIVISION,
  UPPER(OPERATOR) AS OPERATOR,
  ROUND( TOTAL_CONFIRMED/Y.TOTAL_ORDERS * 100,2) AS COMPLETION_PERCENTAGE,
  ROUND(TOTAL_CANCELED/ Y.TOTAL_ORDERS * 100,2) AS CANCELETION_PERCENTAGE
FROM
  Y;
```

DIVISION	OPERATOR	COMPLETION_PERCENTAGE	CANCELETION_PERCENTAGE
RANGPUR	ROBI	79.43	13.65
RANGPUR	GP	78.41	12.06
RANGPUR	AIRTEL	82.15	10.07
RANGPUR	BL	85.99	5.8
KHULNA	GP	80.54	9.25
KHULNA	AIRTEL	84.71	7.85
CHITTAGONG	ROBI	73.3	18.23
CHITTAGONG	AIRTEL	80.11	10.22
CHITTAGONG	GP	78.15	14.01
SYLHET	GP	83.17	10.89
SYLHET	AIRTEL	80.44	12.97
DHAKA	BL	78.68	12.01
RAJSHAHI	GP	80.79	10.63
RAJSHAHI	ROBI	74.38	15.15
RAJSHAHI	BL	78.21	12.18
RAJSHAHI	AIRTEL	79.06	11.8
DHAKA	ROBI	77.07	15.87
DHAKA	AIRTEL	80.07	13.03
DHAKA	GP	75.62	15.35
BARISAL	GP	81.03	9.13
MYMENSINGH	BL	77.74	9.97
KHULNA	ROBI	74.92	16.61
KHULNA	BL	75.43	13.65
SYLHET	BL	71.43	10.39
SYLHET	ROBI	76.72	15.87
BARISAL	ROBI	73.72	17.06
BARISAL	BL	75.49	13.44
MYMENSINGH	AIRTEL	81.18	9.17
BARISAL	AIRTEL	78.71	12.17
MYMENSINGH	GP	78.46	14.03
MYMENSINGH	ROBI	76.22	14.84
CHITTAGONG	BL	75.09	10.57



The percentage of order statuses like completed and canceled varies by operator and division, reflecting differences in service performance and customer satisfaction. For instance, some operators may have higher completion rates in divisions with better logistics. Detailed insights into these percentages can guide improvements in order fulfillment processes.



1. How many products are offered by each operator?
2. Calculate Potential revenue loss due to cancellation

1

```
SELECT
UPPER(OPERATOR) AS OPERATOR,
COUNT(DISTINCT PRODUCT_ID) AS TOTAL_PRODUCTS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
GROUP BY
  1;
```

OPERATOR	TOTAL_PRODUCTS
GP	139
ROBI	172
AIRTEL	192
BL	119

2

```
SELECT
  SUM(SELLING_PRICE) AS TOTAL_REVENUE_LOSS,
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
  ORDER_STATUS = "Cancel";
```

TOTAL_REVENUE_LOSS
1161633



The potential revenue loss due to cancellations is 1161633, which represents the total value lost from canceled orders. This amount highlights the financial impact cancellations have on overall revenue.



Each operator offers a different number of products: GP offers 139 products, ROBI offers 172 products, AIRTEL offers 192 products, and BL offers 119 products. This variation reflects the diversity in product portfolios across operators. The differences may influence customer preferences and sales performance.



Calculate product popularity by validity period.

```
SELECT
  VALIDITY,
  COUNT(ORDER_ID) AS POPULARITY
FROM
  `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
GROUP BY
  1
ORDER BY
  POPULARITY DESC;

-- ADVANCED CODE USING CTE AND WINDOWS FUNCTION

WITH M AS (
  SELECT
    VALIDITY,
    CONCAT(UPPER(OPERATOR), "-", PRODUCT_NAME) AS PRODUCT_NAME,
    COUNT(ORDER_ID) AS POPULARITY,
    ROW_NUMBER() OVER (PARTITION BY VALIDITY ORDER BY COUNT(ORDER_ID) DESC) AS POPULARITY_BY_VALIDITY
  FROM
    `organic-reef-427307-i3.offer_load.Offer_load_Transaction`
  GROUP BY
    VALIDITY, OPERATOR, PRODUCT_NAME
)
SELECT
  VALIDITY,
  PRODUCT_NAME,
  POPULARITY,
  POPULARITY_BY_VALIDITY
FROM
  M
ORDER BY
  POPULARITY DESC;
```

VALIDITY	PRODUCT_NAME	POPULARITY	POPULARITY_BY_VALIDITY
30 days	ROBI-10 GB	1298	1
30 days	ROBI-20 GB	1148	2
30 days	GP-500 min	666	3
30 days	GP-40 GB - 800 min	581	4
30 days	AIRTEL-10 GB	513	5
30 days	ROBI-30 GB	463	6
30 days	GP-50 GB - 1600 min	385	7
30 days	AIRTEL-16 GB - 300 min	382	8
30 days	GP-25 GB - 400 min	356	9
30 days	BL-40 GB	353	10
30 days	GP-40 GB - 750 min	329	11
30 days	AIRTEL-20 GB	309	12
30 days	GP-25 GB	307	13
30 days	ROBI-165 min	290	14
30 days	AIRTEL-40 GB - 200 min	270	15
30 days	ROBI-50 GB - 1600 min	251	16
30 days	GP-35 GB - 800 min	241	17
30 days	GP-1 GB - 1000 min	231	18
30 days	AIRTEL-30 GB	224	19
30 days	AIRTEL-18 GB - 400 min	216	20
30 days	GP-400 min	214	21
30 days	ROBI-40 GB	197	22
30 days	ROBI-30 GB - 700 min	197	23
30 days	AIRTEL-30 GB - 700 min	190	24
30 days	ROBI-45 GB	173	25
30 days	AIRTEL-275 min	166	26
30 days	ROBI-5 GB	154	27
30 days	AIRTEL-55 GB - 1000 min	153	28
30 days	ROBI-15 GB	147	29
30 days	BL-50 GB	144	30
30 days	ROBI-1000 min	137	31
30 days	AIRTEL-50 GB - 1000 min	136	32
30 days	GP-40 GB	135	33



Product popularity can be assessed by analyzing sales or orders based on their validity periods. By grouping products by validity period and measuring metrics like total sales or order counts, we can determine which validity period is most favored by customers.



What are the top 3 or 5 most-selling products or bundles in key divisions like Dhaka and Rangpur?

```
WITH Z AS
(SELECT
  DIVISION,
  PRODUCT_NAME,
  BUNDLE_TYPE,
  COUNT(ORDER_ID) AS TOTAL_SELLING,
  ROW_NUMBER() OVER (PARTITION BY DIVISION ORDER BY COUNT(ORDER_ID) DESC) AS RANKS
FROM
  organic-reef-427307-i3.offer_load.Offer_load_Transaction
WHERE
  DIVISION IN ('Dhaka', 'Rangpur')
GROUP BY
  1,2,3
ORDER BY
  DIVISION, TOTAL_SELLING DESC)

SELECT
  DIVISION,
  PRODUCT_NAME,
  BUNDLE_TYPE,
  Z.TOTAL_SELLING
FROM
  Z
WHERE
  RANKS <= 5;
```



In Dhaka, the top-selling products are 10 GB Internet (486 units), 20 GB Internet (329 units), and 50 GB - 1600 min Internet & Minute bundle (292 units). In Rangpur, the top-selling products are 500 min Minute bundle (113 units), 20 GB Internet (111 units), and 10 GB Internet (110 units). These products dominate sales in their respective divisions.

DIVISION	PRODUCT_NAME	BUNDLE_TYPE	TOTAL_SELLING
Dhaka	10 GB	Internet	486
Dhaka	20 GB	Internet	329
Dhaka	50 GB - 1600 min	Intenet & Minute	292
Dhaka	500 min	Minute	260
Dhaka	40 GB - 800 min	Intenet & Minute	227
Rangpur	500 min	Minute	113
Rangpur	20 GB	Internet	111
Rangpur	10 GB	Internet	110
Rangpur	40 GB - 800 min	Intenet & Minute	104
Rangpur	50 GB - 1600 min	Intenet & Minute	89



The End